Abstract Algebra Term Paper: Finite Fields F_{p^n}

Construction via Irreducible Polynomials and Applications

Elliot Smythe

December 12th, 2025

1 Introduction

Finite Fields are everywhere across Mathematics, Computer Science, Engineering and more. This topic is about the construction of finite fields, as well as the applications these fields have in computer science and digital design. We include various lemmas for a proof of the fundamental theorem of finite fields, and give two examples of the construction of finite fields with irreducible polynomials.

2 Definitions and Lemmas

- Definition: Extension Field
  A field E is an *extension field* if $F \subseteq E$ and the operations of F are those of E restricted to F (Gallian, Chapter 19).

- Definition: Splitting Field
  For an extension field E and $f(x) \in F[x]$ with $deg(f(x)) \geq 1$, we say $f(x)$ *splits* in E if there are elements $a \in F$ and $a_1, a_2, ..., a_n \in E$ s.t. $f(x) = a(x - a_1)(x - a_2)...(x - a_n)$.
  We call E a *splitting field* for $f(x)$ over F if $E = F(a_1, a_2, ..., a_n)$. (Gallian, Chapter 19).

- Lemma 1: The polynomial $f(x) = x^{p^n} - x$ has no repeated roots.

  Proof. First, compute the derivative, $f'(x)$ as $p^n x^{p^n - 1} - 1 = -1 \neq 0$. For $f(x)$ to have repeated roots, $(x - a)$ would need to divide $f(x)$ and $f'(x)$. But since $f'(x) = -1$, this is impossible, and thus, $f(x)$ has no repeated roots. □

- Lemma 2: If F is a field of order p^n, then every element of F is a root of $x^{p^n} - x$.

  Proof. Equivalently: for every $a \in F, a^{p^n} = a$

  To begin, observe that if $|F| = p^n$, then $|F^*| = p^n - 1$, and let $a \in F$. If

$a = 0$, then $0^{p^n} - 0 = 0$ implies that 0 is a root. If $a \neq 0$, then $a \in F^*$. By Lagrange's Theorem, $a^{|F^*|} = a^{p^n - 1} = 1$. This implies that $a^{p^n} = a$, which implies that a is a root of $x^{p^n} - x$.
This is a standard result for field theory; see Gallian Chapter 21. □

- Lemma 3: F^* is cyclic

  Proof. F^* is a finite abelian group because it inherits commutativity from F. From Lemma 2, every element satisfies $x^{p^n - 1} - 1 = 0$, and $x^k = 1$ can have at most k solutions in F. Let $p^n - 1 = m$. The sum of the numbers of elements whose order divides some d, taken over all divisors d of m, must equal m. If no element has order m, then every element satisfies $a^j = 1$ for some $j < m$. This implies that $|F^*| < m$, a contradiction. Therefore, there must exist $a \in F$ s.t. $ord(a) = p^n - 1$. This a generates F^*, and F^* is therefore cyclic.
  See Lidl-Niederreiter, Chapter 1. □

3 The Fundamental Theorem of Finite Fields

This theorem states that every finite field:

- i) has order p^n for a prime p
- ii) is unique up to isomorphism
- iii) can be constructed as a quotient of $F_p[x]$ by an irreducible polynomial of degree n.

This proof follows the standard result found in Gallian and Herstein.

Proof. i) Every finite field has order p^n for a prime p.
Let F be a finite field and let $char(F) = k$. By way of contradiction, assume $k = ab$ for $1 < a, b < k$ in F. Suppose $0 = k$. Then $0 = k \cdot 1 = ab \cdot 1 = (a \cdot 1)(b \cdot 1)$. Fields have no zero divisors, so $a \cdot 1 = 0$ or $b \cdot 1 = 0$, but this is a contradiction of the minimality of the characteristic. Thus, $k \neq ab$, and therefore k is prime. Going forward, let $char(F) = p$. Now take $A = \{0, 1, 1+1, ..., p-1\} \subseteq F$. A is closed under addition and multiplication mod p, and $char(F) = p$ implies that $A \cong Z/pZ$ since $Z/pZ = \{0, 1, 2, ..., p-1\}$. Because $F_p \subseteq F$, treat F as a vector space over F_p. Since F is finite, the dimension of F is finite. Let $n = \dim_{F_p}(F)$, and choose $\{v_1, v_2, ..., v_n\}$ as a basis of F. Every $a \in F$ can be written as $a_1 v_1 + a_2 v_2 + ... + a_n v_n$ with $a_1 \in F_p$. In this linear combination, there are p slots for each a_i to fill. Therefore, $|F| = p^n$.

□

Proof. ii) By Lemma 2, every element of F is a root of $x^{p^n} - x$, so this polynomial has at least p^n roots in F. But $deg(x^{p^n} - x) = p^n$, so $x^{p^n} - x$ has exactly p^n roots. This means that these roots make up F entirely. Define $f'(x)$ as the

derivative of f. For $f(x) = (x^{p^n} - x)$, $f'(x) = -1 \neq 0$. Thus, by Lemma 1, $x^{p^n} - x$ has no repeated roots, which means its roots are distinct. Now, let F_p be the prime subfield of F. Since $x^{p^n} - x$ splits over F, the splitting field K of $x^{p^n} - x$ over F_p is inside F ($K \subseteq F$). But K contains all roots of $x^{p^n} - x$, and these roots are the elements of F, so $F \subseteq K$. Therefore, $F = K$, and F is the splitting field of $x^{p^n} - x$ over F_p. The above logic works for any splitting field K of order p^n over F, and one can draw an isomorphism between splitting fields of the same base field and and same polynomial. Therefore, F is isomorphically unique. □

Proof. iii) F is an n-dimensional vector space over F_p, so $[F : F_p] = n$. Take $F^* = F \setminus \{0\}$ (from Lemma 2), viewed as a multiplicative group. This is cyclic of order $p^n - 1$ by Lemma 3 (and finite). Choose a s.t. $F^* = \langle a \rangle$. Now observe $F_p(a)$: if $a \in F_p(a)$, then $a^k \in F_p(a)$ for all k. Thus, every non-zero element of F lies in $F_p(a)$. Additionally, 0 is in every subfield by definition. Therefore, $F = F_p(a)$, and F is an extension of F_p. Let $f(x)$ be the minimal polynomial of a, a monic of minimal degree. By way of contradiction, assume $f(x) = g(x)h(x)$ with $deg(g), deg(h) < deg(f)$. Suppose $0 = f(a) = g(a)h(a)$. Because F is a field with no zero divisors, $g(a) = 0$ or $h(a) = 0$, a contradiction of minimality of f(x). Thus, $f(x) \neq g(x)h(x)$, so f is irreducible. Consider $B = \{1, a, a^2, ..., a^{k-1}\}$ with $k = deg(f(x))$. These elements span $F_p(a)$ (and are linearly independent because there would have to be a non-zero polynomial of degree less than k if they were dependent). Thus, B forms a basis, and $[F_p(a) : F_p] = k$. Because $F = F_p(a)$ and $[F : F_p] = n$, the degree of $f(x)$ is n (and we have $k = n$). Now define $\phi : F_p[x] \to F$ as $\phi(g(x)) = g(a)$. ϕ is a ring homomorphism, and the image of ϕ is $F_p[a] = F_p(a) = F$ implies that ϕ is surjective. The kernel of ϕ is $(f(x))$ (the ideal) because $f(x)$ generates all polynomials that have a as a root, and if a is a root, it maps to 0. Finally, by the First Isomorphism Theorem, $F_p[x]/f(x) \cong F$ Thus, we have an irreducible $f(x)$ with degree n that can construct F as a quotient of $F_p[x]$. □

4 Example 1: Simple Construction

We now work a simple example. Take $F_2 = \{0, 1\}$ and choose $f(x) = x^2 + x + 1$. With this, $f(0) = 1$ and $f(1) = 3 = 1 mod 2$. Therefore, there are no zeroes, and since this polynomial is a quadratic, this polynomial is irreducible in F_2. Now, take $F_4 = F_2[x]/(x^2 + x + 1)$.

- Working in modulo $x^2 + x + 1$, $x^2 = x + 1$.
- The elements of F_4 with degree less than 2 are $\{0, 1, x, x + 1\}$.
- Define addition as $(x+1)+x = 1$ and multiplication as $x \cdot x = x^2 = x+1$.
- Check Inverses: $1 \cdot 1 = 1$, so 1 is its own inverse. $x(x + 1) = x^2 + x = x + 1 + x = 1$, so x and $x + 1$ are inverses of each other. Thus, every non-zero element has an inverse. Therefore this is a field, constructed from an irreducible polynomial.

5 Example 2: Digital Design

Computer hardware depends on high and low voltages, which are abstracted to mean 0s and 1s in digital design. A bit is an element that can take either value. To implement more complex systems, computer scientists implement logic gates and various representations of strings of bits (machine code, bytes, hexadecimal, etc). These can both be thought of in terms of fields.

5.1 Logic Gates:

Take the field $F_2 = \{0, 1\}$ with integer addition and multiplication mod 2. Testing out all combinations of elements, the operations of F_2 look like the following:

Addition
$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 0$

Multiplication
$0 \cdot 0 = 0$
$0 \cdot 1 = 0$
$1 \cdot 0 = 0$
$1 \cdot 1 = 1$

And the truth table for a XOR gate and AND gate look like the following:

A	**B**	**XOR**
0	0	0
0	1	1
1	0	1
1	1	0

A	**B**	**AND**
0	0	0
0	1	0
1	0	0
1	1	1

If one defined a map between these tables, element by element, they would be isomorphically equivalent.

5.2 Bit Grouping:

For the ease of calculation, we take 4-Bit groupings and try to define operations similar to the single bit examples above. The set $\{0, 1\}^4 = \{x_1, x_2, x_3, x_4 : x_i \in F_2\}$ is the immediate candidate for use, and it does work for component wise addition (analogous to the XOR operation). For example, the standard base-2 operation $1000 + 0001 = 1001$ is the same as the component wise $(1, 0, 0, 0) + (0, 0, 0, 1) = (1, 0, 0, 1)$. However, this interpretation runs into problems if it is necessary to find a function with an inverse, which would be needed for a lookup table or digital register. The solution to this is to use polynomial multiplication modulo some irreducible polynomial.
Say we need to create a 4-bit lookup table (LUT). To do this, assume we work in $F_{2^4} = F_2[x]/(x^4 + x + 1)$. Now to check the value this LUT would provide, verify the inverse of x^2+x+1: $(x^2+x+1)(x^2+x) = x^4+x^3+x^2+x^3+x^2+x = x^4 + x = x + 1 + x = 1$). Now we could implement this, and every time our

design receives the digital input 0111, it would output 0110, because $x^2 + x + 1$ corresponds to 0111 and $(x^2 + x)$ corresponds to 0110.
Multiplication by inverses creates a bijection for non-zero elements, which allows us to create non-linear, invertible functions for look up tables that are very useful for constructs like AES S-Boxes (which is defined using F_{2^8}, but the arithmetic of this field is messier than the above example). See Paar & Pelzl.

6 Remarks and References

6.1 Remarks

- The set of AND and XOR gates is universal, meaning all common logical gates can be implemented with a combination of these two gates and constant boolean values. We will not delve into this subject but it is very useful for digital design.
- The most significant bit is usually defined as the leftmost bit in computer architecture, but for consistency with mathematical notation, I assigned the MSB to the rightmost element in section 5.2. It does not cause a large difference.
- In section 2, we discuss a bit of calculus to be used in later proofs. I will take this topic to be known, as formal discussions of the derivative are reserved for other classes and are beyond the scope of a paper with a page limitation.
- In section 3, we use known facts and techniques about isomorphisms, splitting fields, and particularly, vector spaces. We take these to be known for the same reasons as above.

6.2 References

- J. Gallian, *Contemporary Abstract Algebra*, Chapters 18, 19, 20, and 21
- I.N. Herstein, *Topics in Algebra*, Section on Extension Fields
- R. Lidl and H. Niederreiter, *Finite Fields*, Chapters 1 and 2.
- Harris & Harris, *Digital Design and Computer Architecture, RISC-V Edition* (general context)
- Paar & Pelzl, *NIST FIPS 197, Advanced Encryption Standard*, Section 4